

# Socio Economic Factors Effecting Employability in England and Wales

## 1 Abstract

The report is focused on what impact socio-economic factors have on the labour market. The data has been drawn from 2011 and 2021 UK Census data(3). The focus has been to understand the economic disparity among two sexes, how employment, unemployment rates have changed over the years and, employability and educational qualification in an area go hand in hand.

## 2 Introduction

A lot of socio and economic factors have their effect on the labour market. A lot of social features play a vital role in shaping market condition. Here, I have tried to understand few of these features. From decades, economic activity has been measured with gender. Here, I have tried to draw a conclusion how much gender gap exists when it comes to employability. I have tried to find answers to questions like do traditional gender roles are still relevant in today's market. Results which are backed with data help us to make solid conclusions.

Another aspect covered here is Educational Qualification. We all know that educational qualification plays a vital role in getting a job. I have attempted to get insights about educational qualifications people had in 2011 and if anything has changed in 2021. This analysis will pave way to the final part where I have calculated employment and unemployment rates. These two parameters give a clear picture of the market condition. It further facilitates in predicting future rates.

## 3 Task Definition

Tamara Munzner divides the process of visualization into 3 steps:

1. Data Abstraction – What type of data are you using?
2. Task Abstraction – Why is the user looking at it?
3. Idiom – How do you show the data, based on the data type and tasks?(4)

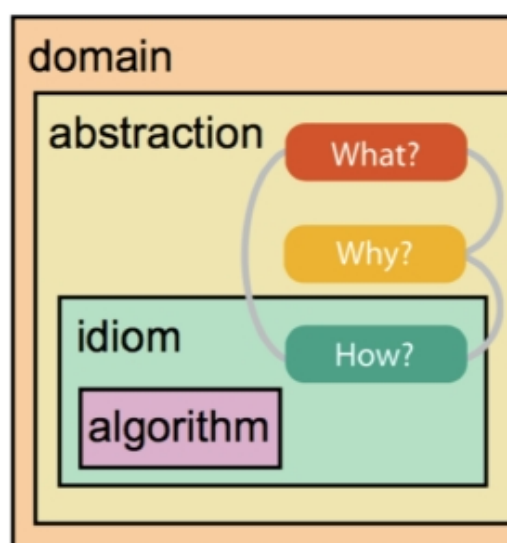


Figure 1: Muzner's Task Taxonomy

- In this task we have used tabular data from UK Census 2021 and 2011. For the first classification, dataset from 2011 census have been taken which describes economic activity of people grouped by Male and Females.

The second analysis is based educational qualifications. Two tables have been taken from 2011 and 2021 respectively. The data set describes different qualification levels spread across 10 regions of England and Wales. Finally I have calculated employment rate and unemployment rate from the tables of economic activities for both years which are spread across all the Local District Areas. Further, I have discussed how the data has been categorised into items and described the attributes.

- The second process is task abstraction. This part mainly answers 'why' is the user looking at it? The main focus here is to draw insights from the tables which turn out to be important factors in analyzing the market trends.
- The final step is 'idiom' or which visual encoding suits the task. For each and every task done, I have discussed about Visualisation Justification which discusses about the visualisation, why was it chosen.

## 4 Economic Activity Status based on Sex

### 4.1 Objective:

The analysis done here is based on the economic activity status of people living in England and Wales. The data has been drawn from the 2011 Census. The analysis will dive deep into understanding the labour market spread across England and Wales and how it varies by gender. In this section, I will examine the distribution of employment, unemployment and economic inactivity across males and females. Key insights that I tried to draw through this analysis are:

- Compare male vs. female participation in each category of economic activity.
- Identify which economic activity shows the greatest gender imbalance
- Spot outliers like a greater population in a particular economic activity.

In the next section of data preparation and abstraction, I will explain about the data cleaning steps done.

### 4.2 Data Preparation and Abstraction:

The data has been collected from the table *DC6203EW*. The table originally had 20 columns. Since it is very difficult to make analysis with so many columns I merged and redefined the structure of the table having only three columns. I created three columns of Sex, Economic activity, and Count. In the column of Economic activity I merged the important topics (*Economically active: Total*, *In employment: Total*, *Employee: Total*, *Unemployed: Total*, *Economically inactive: Total*, *Looking after home or family*) on which I wanted to do analysis. The raw data also had extra texts which had to be eliminated. These data cleaning steps helped to achieve this table. The key data types in "Economic Activity based on Sex":

- Categorical: Sex: "Males" vs. "Females" Economic Status: e.g. "In employment," "Unemployed," "Looking after home or family," etc.
- Quantitative: Count: number of persons of each sex in each category of economic status.

### 4.3 Analysis on Economic Activity Status:

1. During the census of 2011, approximately 28.81 million usual residents aged 16 years and over in England and Wales were economically active (63.34%) while 16.68 million (36.66%) were economically inactive.
2. To further understand the market distribution, 26.68 million people were in employment. Among which 52.88% were male employees while approximately 47.07% were female work force.
3. Almost 16.68 million people which approximates to 36.66% of the total population in this period were economically inactive. Many reasons have been taken into account in the census data for the inactivity like retirement, long-term sick or disabled or looking after home. I took into account for the share of population who were

looking after home or family. Approximately 1.63 million females which is 9.77% of the total population were economically inactive - looking after home and family. This turned out to be a greater portion of population as compared to men who were inactive due to the same reason.

|    | Sex     | Economic Status              | Count       |
|----|---------|------------------------------|-------------|
| 0  | Males   | Economically active: Total   | 1,53,65,658 |
| 1  | Females | Economically active: Total   | 1,34,52,697 |
| 2  | Males   | In employment: Total         | 1,41,16,119 |
| 3  | Females | In employment: Total         | 1,25,65,449 |
| 4  | Males   | Employee: Total              | 1,07,85,008 |
| 5  | Females | Employee: Total              | 1,07,74,894 |
| 6  | Males   | Unemployed: Total            | 12,49,539   |
| 7  | Females | Unemployed: Total            | 8,87,248    |
| 8  | Males   | Economically inactive: Total | 67,90,356   |
| 9  | Females | Economically inactive: Total | 98,88,069   |
| 10 | Males   | Looking after home or family | 1,68,490    |
| 11 | Females | Looking after home or family | 16,28,030   |

Figure 2: Economic activity based on sex

#### 4.4 Visualisation Justification:

The dashboard combines a coloured table and a stacked bar chart which shows 2011 Census economic activity based on sex. First at the top is heat-map based table which lists Males and Females across different economic activities. Each cell is shaded from orange to brown. The darker hues signal larger population count which is overlaid with figures. The legend helps us understand how the tone changes with increase in population.

Combining both the aspects of heat map and actual figures, viewers can understand which economic activity has larger portion of population and actual figures give the idea of exact numbers which can be used for any further analysis.

The stacked bar chart echoes the same data but gives a better visual representation of the difference in participation of males in females across different economic activities. The orange bars represent male population and the blue bars represent the female population. The population axis is marked in millions. The data types economic activities and sex have been arranged alphabetically.

## 5 Highest Level of Qualification

### 5.1 Objective:

The analysis done here is based on the level of qualification people have. I have taken into account 10 different regions across England and Wales and sought insight on different qualification levels. The distribution of varied educational levels helps us understand the labour market, as education is directly related to the employability of a person. The key insights that I tried to draw through this analysis are:

- Spread of the educational level in these 10 regions of England and Wales.
- Analyse Change in the qualification levels from 2011 to 2021.

- Identify outliers: Which area has an abnormal level of educational qualification throughout these years.

## 5.2 Data Preparation and Abstraction:

The census data for qualification levels have been collected from two datasets. Table *QS501EW* has been taken from census of 2011 and table *TS067* from the census of 2020. Both the tables have similar kind of data types. The data sets have 9 categorical data types: 'region', 'Total: All usual residents aged 16 years and over', 'No qualifications', 'Level 1 and entry level qualifications', 'Level 2 qualifications', 'Apprenticeship', 'Level 3 qualifications', 'Level 4 qualifications or above', 'Other qualifications'.

Similar kind of preprocessing steps have been applied to both tables. The column names have been renamed to have uniformity across both tables. a lot of meta-data and rows having NaN values have been removed. The aim is to combine both the tables and another column has been added signifying the year.

These preprocessing steps have helped to create a combined table for highest level of qualification spread across 10 regions for years 2011 and 2021 which has been used for dimensionality reduction.

## 5.3 Analysis on Qualification Level:

The basic analysis that can be drawn is the level of education in all regions. Educational qualification plays one of the key factors in determining the employability of a person. If we see the region of London, it had around 2.4 million having Level 4 educational level. This number significantly grew to 3.3 million in 2021. But what didnot change was the population of people having no qualification. It was approximately 1.1 million in 2011 and remained same in 2021. This indicates that though people have started pursuing higher qualifications, there is still a section of the society which is marginalized and deprived of education.

On the other hand if we talk about the region North West, it has around 1.4 million people who have no qualification. This is the highest among all the regions. Places like Yorkshire and West Midlands also share a high population of people having no qualification.

### 5.3.1 Dimentionality Reducation:

As I combined 2011 and 2021 census counts for seven qualification categories across England & Wales' ten standard regions, I applied PCA to to extract two principal axes (PC1, PC2). These captured 76.2 % and 18.0 % of the total variance, respectively.

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=== PCA Loadings ===

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|                        | PC1 loading | PC2 loading |
|------------------------|-------------|-------------|
| No qualifications      | -0.470578   | -0.155306   |
| Level 1 qualifications | -0.255347   | -0.223693   |
| Level 2 qualifications | -0.188661   | 0.056857    |
| Apprenticeship         | 0.014507    | 0.354700    |
| Level 3 qualifications | 0.146889    | 0.714519    |
| Level 4 qualifications | 0.808062    | -0.313548   |
| Other qualifications   | -0.054873   | -0.433529   |

Figure 3: PCA score for 7 qualification levels

- High PC1 scores therefore identify regions with unusually large shares of degree-holders.
- PC1 is dominated by Level 4+
- PC2 contrasts Level 3 and Apprenticeship. Thus PC2 highlights regions that are especially strong (or weak) in mid-level credentials.

=== PCA Scores ===

|    | region                   | Year | PC1       | PC2       |
|----|--------------------------|------|-----------|-----------|
| 0  | South West               | 2011 | -0.026560 | -0.011440 |
| 1  | Wales                    | 2011 | -0.073710 | -0.017247 |
| 2  | East                     | 2011 | -0.053536 | -0.027474 |
| 3  | East Midlands            | 2011 | -0.076148 | -0.013223 |
| 4  | North West               | 2011 | -0.068905 | -0.012100 |
| 5  | North East               | 2011 | -0.094551 | -0.001897 |
| 6  | South East               | 2011 | 0.001858  | -0.025463 |
| 7  | West Midlands            | 2011 | -0.087356 | -0.022419 |
| 8  | Yorkshire and The Humber | 2011 | -0.082321 | -0.011946 |
| 9  | London                   | 2011 | 0.080107  | -0.087953 |
| 10 | South East               | 2021 | 0.088055  | 0.018259  |
| 11 | South West               | 2021 | 0.065923  | 0.037916  |
| 12 | Wales                    | 2021 | 0.034060  | 0.027784  |
| 13 | West Midlands            | 2021 | 0.007902  | 0.025358  |
| 14 | Yorkshire and The Humber | 2021 | 0.012677  | 0.033133  |
| 15 | North West               | 2021 | 0.033129  | 0.031362  |
| 16 | North East               | 2021 | 0.009295  | 0.048304  |
| 17 | London                   | 2021 | 0.178563  | -0.053033 |
| 18 | East Midlands            | 2021 | 0.014722  | 0.040388  |
| 19 | East                     | 2021 | 0.036796  | 0.021693  |

Figure 4: Regional Scores

- London (2021) has the highest PC1 score (+0.18)—far to the right of all others—signals its exceptionally large share of degree-holders. It has a negative PC2 (−0.05) shows London is slightly below average on mid-level (Level 3) qualifications.
- Other High-PC1 Regions like South East sits to the right of the vertical zero-line, indicating above-average proportions of Level 4+.
- Places like North East, East Midlands in 2011 appear far left, denoting higher shares of residents with no qualifications and fewer degree-holders.
- **Rightward Movement on PC1:** Every region has drifted rightward over the decade, showing that degree-level attainment grew everywhere.
- **Vertical (PC2) Shifts:** Regions like East Midlands and Yorkshire & The Humber have moved upward, reflecting expansion in Level 3/apprenticeship pathways; while, London has moved slightly downward, reinforcing its focus on Level 4/university level qualifications.
- PC1's 76% explains that the variance highlights the biggest change and regional differentiation which is in degree-level attainment vs. no-qualification shares.

## 5.4 Visualisation Justification:

I have created two dashboards to completely capture the insights that I have drawn from these two tables.

In the first dashboard has two bar charts stacked one upon another for the years 2021 and 2011 respectively. These bar charts clearly show the distribution of population across these 10 areas. We can easily have a view of different qualification levels people have in an area and also compare it to the corresponding year. This helps us to understand if there has been any significant change qualification levels in a decade. The table has a lot of attributes but for the ease of analysis, I have taken into account four levels: 'Level 1 qualification', 'Level 4 qualification', 'Apprenticeship' and 'No qualifications'. These columns show the right amount of data for us to understand what level of educational qualification people hold in an area.

- Bar graphs are not able to give information about the exact numbers but they represent comparisons very well. Bar length on a common baseline. According to Cleveland & McGill (1984)(1), people judge lengths far more precisely than color shades or angles. This makes it easy to compare For example, if we take London. It

has highest number of people with Level 4 qualification among all the areas. Also, there has been a jump in population having Level 4 qualification from year 2011 to 2021.

- I have used qualitative palette like green, blue, yellow and teal to be mutually distinct and color-blind-friendly. I have used hues to merely distinguish categories. It has been not used to encode quantity.
- I have arranged all ten regions along a single shared axis rather than separate panels. This shared-scale approach leverages smarter comparisons across regions.
- Hovering over to each bar graph gives the exact count of population.
- By keeping the design uncluttered and leveraging humans' ability to compare lengths along a shared axis, I aimed to minimize mental arithmetic.
- For this visualization, the choice of bar graph is better than other kind of graphs like pie charts or heatmap. Pie charts compromise positional accuracy. Heatmaps can show proportions but would require us to interpret magnitude, which is less precise than bar length.

The second dashboard captures clusters of regions which have similar kind of qualifications and how it has changed from 2011 to 2021. This has been achieved by applying dimensionality reduction technique - PCA. It is an unsupervised way to compress seven correlated qualification shares into two dimensions for easy visualization.

- By plotting regions on PC1 vs. PC2, I have leveraged the ability to perceive spatial clusters and detect outliers. Through the scatter plot, it is easy to detect London's outlier status (far right).
- By summarizing seven qualification categories into two axes, the scatter plot gives a clear, cognitive and efficient overview of how England & Wales' educational profile has evolved regionally between the two censuses.

## 6 Employment Rate and Unemployment Rate across England and Wales

### 6.1 Objective:

The final analysis I have done here is by comparing the employment and unemployment rates in the two census years. This will give us clear statistics to understand how the labour market has changed in a decade. Key insights that I have drawn are:

- Employment and Unemployment Rate across all the districts in two district years.
- Visualise a clear comparison to show the change of rate
- Predict employment and unemployment rate for the year 2031.

### 6.2 Data Preparation and Abstraction:

I have taken two tables - WD601EW and TS066 from the year 2011 and 2021 respectively, describing economic activity status across all districts of England and Wales. Both the tables have same column names and similar kind of data processing steps have been applied as done before. After the basic preprocessing of each table, I got two tables with 10 columns each.

Categorical: 'region', 'All categories: Economic activity', 'Economically active: Employee: Part-time', 'Economically active: Employee: Full-time', 'Economically active: Self-employed with employees: Part-time', 'Economically active: Self-employed with employees: Full-time', 'Economically active: Self-employed without employees: Part-time', 'Economically active: Self-employed without employees: Full-time', 'Economically active: Unemployed', 'Economically inactive: Total'

Next, the aim is to calculate the employment rate and unemployment rate. For this I summed up all the columns which were under Economically active to calculate the employment rate. After calculating the *employment rate and unemployment rate*, I created a new table.

### 6.3 Analysis on Employment rate and Unemployment Rate:

#### 6.3.1 Choropleth Maps of Employment and Unemployment Rates in England & Wales: 2011 vs 2021

I used Choropleth Maps to visualise the employment and unemployment rates across England and Wales. I created two dashboards, for unemployment and employment rates. In each dashboard I have shown two choropleth maps of 2011 and 2021 respectively. This has helped to draw comparisons very effectively.

- Certain post-industrial towns in the North East like Middlesbrough, Hartlepool and South of Wales stand out in darker hues in both years.
- Places like London show exceptionally high employment rate in 2011 and increased further in 2021.

#### 6.3.2 t-SNE analysis of Local Authority Labour Market profiles, 2011 vs 2021

To explore how regional market in England and Wales have changed between 2011 and 2021 censuses, I have computed two simple metrics for each Local Authority District (LAD) : Employment rate (%) and Unemployment rate (%). Then I applied t-SNE to visualize all the points in a single two-dimensional plot.

t-SNE is a non-linear dimensionality-reduction algorithm that is designed to take points from a high-dimensional space and place them in a two-dimensional map. In this case each LAD is a point in  $R^2$  which is defined by employment and unemployment rate and then is placed in a two-dimensional map so that relationships are preserved (like which districts are most similar to each other). Unlike linear dimensionality technique PCA which was applied earlier, t-SNE can reveal clusters of similar points even when those clusters lie on curved manifolds.

I chose t-SNE because I expected few districts would share similar labour-market balances and t-SNE exposed such clusters very well. Another reason was to take into account non-linear shifts. If 2011 data differs from 2021, t-SNE can capture the configuration without assuming linearity. Finally, the resulting 2D scatter naturally separates clusters into islands, making interpretation and labeling straightforward.

#### 6.3.3 Implementation Details:

I used Scikit-Learn's TSNE( $n\_components=2$ ,  $perplexity=10$ ,  $random\_state=42$ ). Perplexity demonstrates 'expected local neighborhood size'. The algorithm then returned two coordinates (tSNE1, tSNE2) for each region and year. I then plotted the point of 2011 in blue and that of 2021 in orange.

#### 6.3.4 Result and Interpretation:

- In a t-SNE scatter plot, neither of the axes correspond to original values. Instead they correspond to coordinates whose purpose is to place similar points together.
- tSNE1 (horizontal axis): I interpretation I could draw is that the single source of variation was how high a region's total employment rate was. t-SNE placed the low-employment areas on the left and high-employment ones on the right. This can be explained using an example from the plot. Region like Babergh which ended up on the left part of the axis has a t-SNE value of -111 and has an employment rate of around 60% in 2011. Whereas region like Forest Heath has employment rate near 90% in 2021 and is placed on the right end of the axis (+ve t-SNE1). Since t-SNE mapping is not linear, we cannot quantify the values but can say that -ve t-SNE1 means lower employment and vice versa.
- tSNE2 (vertical axis): After accounting for overall participation, the next biggest source of difference between councils was how that participation was composed—full-time vs part-time vs self-employed vs unemployed. The algorithm uses tSNE2 to pull apart clusters that have similar totals but different mixes. Two councils with nearly identical employment rate but different unemployment rate will sit at the same horizontal position but up/down along tSNE2.
- The blue dots represent year 2011 and orange dots represent the year 2021. The blue dots are spread from bottom left till around the center. The orange dots sit in the similar loop space but displaced far to the right. Bottom-Left Clusters show regions with low employment and higher inactivity. In 2021, these dots moved to the right of t-SNE1 scale indicating a good improvement in employment.

- The central and the upper arch having both blue and orange dots indicate higher unemployment and maybe a larger section of the population is not in the employability category. The orange dots have moved to the right which means that the economy in these regions have seen gain in 2021.
- Far right are outliers which are all orange dots. Primarily areas like Greater London, Forest Heath and a handful of exceptionally high-employment areas by 2021 are represented.

#### 6.4 Conclusion:

t-SNE plot tells us two aspects of the economy. Every region improved its employment share from 2011 to 2021 (which is the horizontal shift) yet the clusters of low participation, high-employment zones remained profoundly stable over that decade.

#### 6.5 Visualisation Justification:

- The reason for using choropleth maps was to compare the same geographic units across two time-points.
- Placing 2011 and 2021 maps next to each other uses the Gestalt principle of continuity(2) which means that our eyes naturally scan from left to right—and makes it easy to spot where colours lighten or darken over time.
- The legends used in both the dashboards is similar, but the maps have opposite colour ramps showing the difference and the same mid-tone on both maps, make direct comparison valid.
- In the unemployment rate dashboard, I have used warm to cold tones.
- In the employment rate dashboard, I have used cold to warm tones, signifying improvement in the rate.
- Crisp boundaries around Local Authorities guide the eye to the precise shape and location of each unit.
- Hovering over each area, it shows the rate and also a bar graph showing comparison of unemployment and employment rate of that region in that particular year. So it actually gives a deeper prospect of market condition. This helps to gain quick insights.

#### 6.6 Predicting Employment & Unemployment Rates

I have implemented a '*Bayesian logit-linear regression model, estimated via Markov Chain Monte Carlo*' to forecast employment and unemployment trends in 2031.

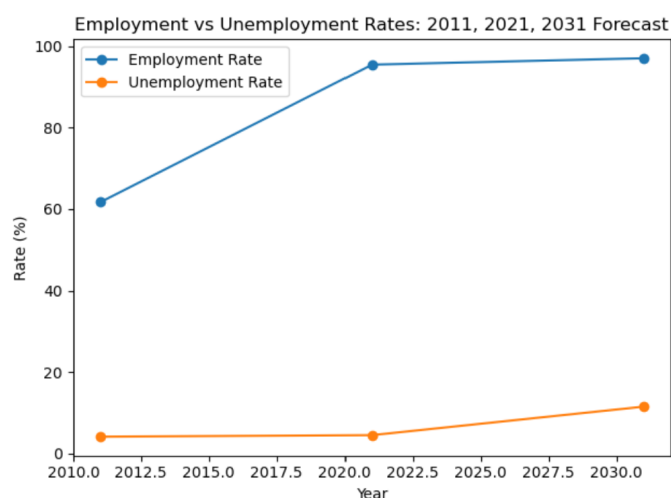


Figure 5: Employment & Unemployment predicted Rates

| Year | Employment Rate | Unemployment Rate |
|------|-----------------|-------------------|
| 2011 | 61.71           | 4.15              |
| 2021 | 95.46           | 4.54              |
| 2031 | 97.02           | 11.52             |

Figure 6: Prediction Table

Employment and unemployment rates live between 0% and 100%. A straight linear regression on those percentages can predict values beyond this range.

## 7 Evaluation

During the process of evaluation, a group of 4 students evaluated my dashboards and answered all the presented questions. They gave feedback about the visual appeal of the dashboards and answered questions. The intention was to create a simple yet information dashboard. They appreciated the concept behind the analysis and visualizations but gave few feedback regarding the use of colours. One of them suggested to apply uniform colour gradient to bar charts of economic activity. I incorporated this which enhanced the visual appeal. Another suggestion during the evaluation that the graphs could be made more informative by adding information. Therefore in the choropleth maps, I added the bar graph of employment and unemployment rates. This doesnot only give precis information just on that topic but also shows the overall picture. One of them suggested me to join few more tables but I felt that though it would give more information but it would also complicate the visualisations. Overcomplicated visualisations are not easy to read or understand. My understanding is that if a visualisation shows too much information then it becomes difficult for the readers to absorb the actual meaning of it.

## 8 Conclusion

Now that the analysis is completed, I have drawn conclusion on the socio economic factors that effect the labour market.

In 2011 16.68 million people aged 16 and above were economically inactive. This constitutes to 36.66% of the total population. Further digging deeper I came across that 9.77% of the total population were doing unpaid labour like taking care of family which constituted mostly of females. This a large chunk of population away from job market, specifically effecting a particular gender.

The data highlighted that traditional gender roles are still prevalent in the society which makes a void in the market. If this percentage of working population takes part in employability, then it can boost the economic condition of homes and that of the country.

The next analysis of qualification level showed us the trends in educational levels. I could conclude that the qualification levels grew from 2011 to 2021 and there were few areas that performed very well as compared to others. There was a pattern in this and it was studied very well by PCA. So we can say that places which economically active tend to show higher educational qualification which in turn boosts the employability. So it is an interconnected factor. Finally I calculated the unemployment and employment rates for both the years. The shift in employment rate was directly proportional to the growth in qualification levels.

Though the conclusion from each table, I have discussed individually but overall I could say that a number of factors effect the employability of an area. An area which is economically very active has better opportunity to gain higher qualifications. So as the market performs better in that area, people continue to enhance their education. But it can also be concluded that people who had no qualification did not drastically decrease in 2021. So, it can be concluded that as population grew, more people joined the work force but a background of people who were devoid of education in 2011 continued to be in that state in 2021. The change in this aspect has not been enough.

The Choropleth mapping has helped in analysing the employment and unemployment rates and draw comparison between both the years keeping the maps side by side. As discussed before, London has shown high rate of employment in both the years. It also showed higher qualification. Definitely these features help it to become the economic hub of entire Europe. Finally using Bayesian logit-linear regression model, I predicted the future employment and unemployment rates. By applying this algorithm, I worked end-to-end on the Muzner's task Taxonomy.

## References

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- [2] Gestalt principle of continuity - Tufte 1990 Available: <https://www.geoff-hart.com/articles/2012/gestalt.html>
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